

CLOSED-LOOP CONTROL · VISION + MOTION

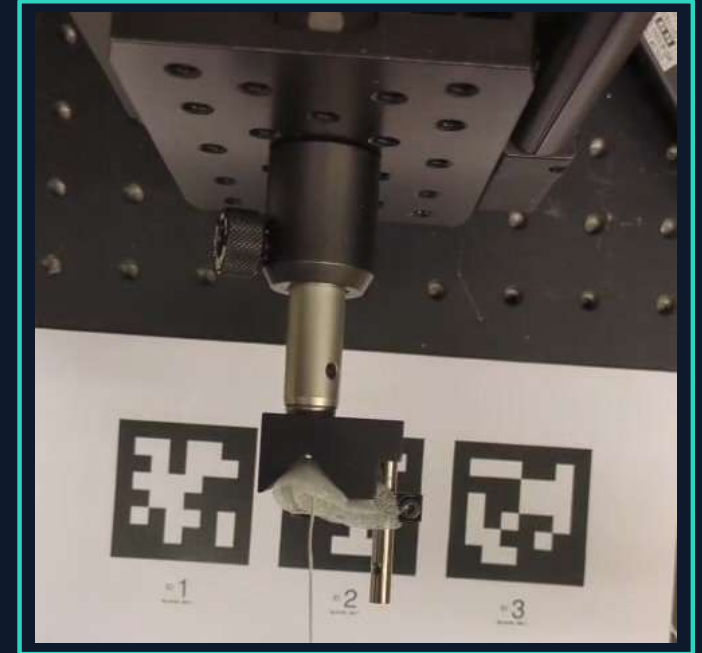
# Visual-Servoing Controller

Centering an AprilTag under an endoscope camera by driving three Thorlabs motorized linear stages over Ethernet.

AprilTag detection

PI controller

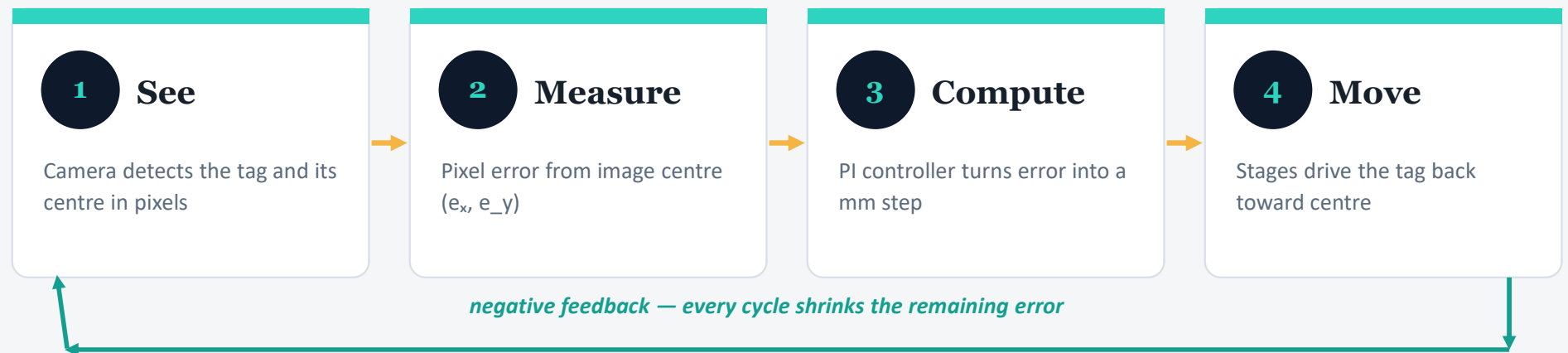
KDC101 stages



Jasper Blank June 2026

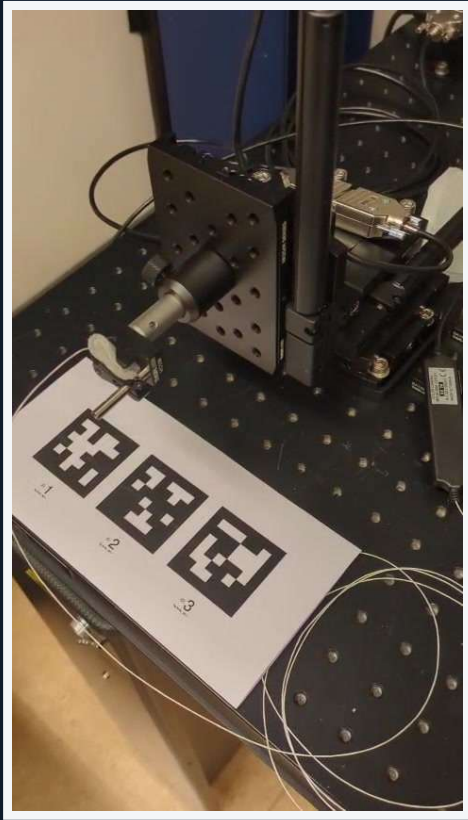
## THE GOAL

# Keep the target centered, automatically



## THE SETUP

# What's on the bench



*Endoscope camera on a 3-axis stage stack, viewing a printed AprilTag sheet (IDs 1-2-3).*

### Endoscope camera

Looks straight down at the target sheet and streams frames to OpenCV for AprilTag detection.

### 3 × KDC101 linear stages

Thorlabs DC-servo motorized actuators that physically move the camera in millimetres along each axis.

### AprilTag target

Printed 36h11 tags; ID 2 is the one the loop keeps centred.

### KCube Ethernet hub

Single hub at 192.168.0.200 — each axis reached on its own TCP port.

WHAT I DO WITH THE LINEAR STAGES

# Three motorized axes

Each Thorlabs KDC101 servo drives one linear stage. Python sends Thorlabs APT binary commands directly to the stage's TCP port — relative or absolute moves, in millimetres, bounded by software limits.

## AXIS 1 *vertical*

Moves the tag UP / DOWN in the image

corrects  $e_y$

## AXIS 2 *horizontal*

Moves the tag LEFT / RIGHT in the image

corrects  $e_x$

## AXIS 3 *depth / spare*

Available; not used by the centering loop

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### RANGE

0–20 mm soft limits, per axis

### MOVE CHECK

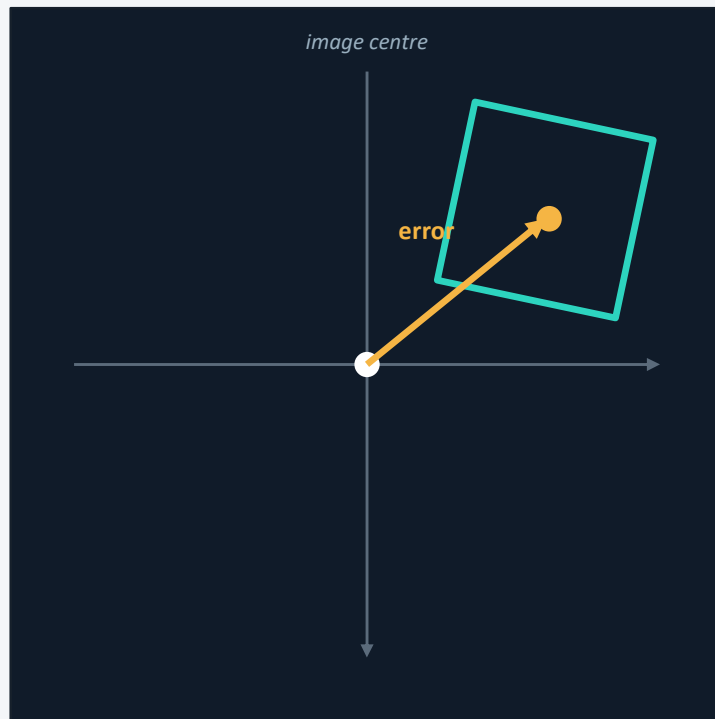
Multi-axis moves validated before any motor moves

### MAX STEP

5 mm per correction cycle

## STEP 1 · SEEING THE TAG

# Measuring the pixel error



Live preview: centre cross · tag outline · tag ID · pixel error · angle

- ▶ **Detector:** OpenCV cv2.aruco reads 36h11 AprilTags in every frame — no extra library needed.
- ▶ **Reports:** tag ID, tag centre in pixels, error vs image centre, and the tag's 2-D angle.
- ▶ **Tracks ID 2** by default — the tag the loop keeps centred.

### PIXEL ERROR

$$\text{error\_x} = \text{tag\_x} - \text{centre\_x}$$

$$\text{error\_y} = \text{tag\_y} - \text{centre\_y}$$

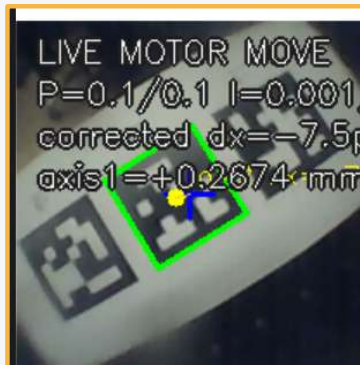
*+x → tag is right of centre · +y → tag is below centre*

## STEP 2 · HANDLING CAMERA ROLL

# Rotating the image so the axes line up

The endoscope can be rolled during production, so the camera's x/y no longer match the stage axes. The code measures the tag's angle and **de-rotates the error before turning it into motor motion**.

RAW — image rolled

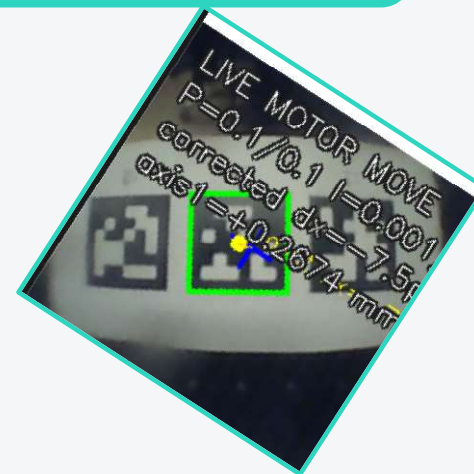


*image axes ≠ stage axes*

by tag angle

de-rotate

COMPENSATED — axes aligned



vertical → axis 1 · horizontal → axis 2

### STEP 3 · THE CONTROLLER

# PI controller

$$\text{motor\_step} = -(P \cdot \text{corrected\_error} + I \cdot \text{accumulated\_error})$$

*the minus sign is the negative feedback that drives the tag toward centre*

P

**0.1 mm / pixel**

proportional gain, both axes

I

**0.001**

integral gain ( = P / 100 )

MAX STEP

**5.0 mm**

clamp per correction cycle

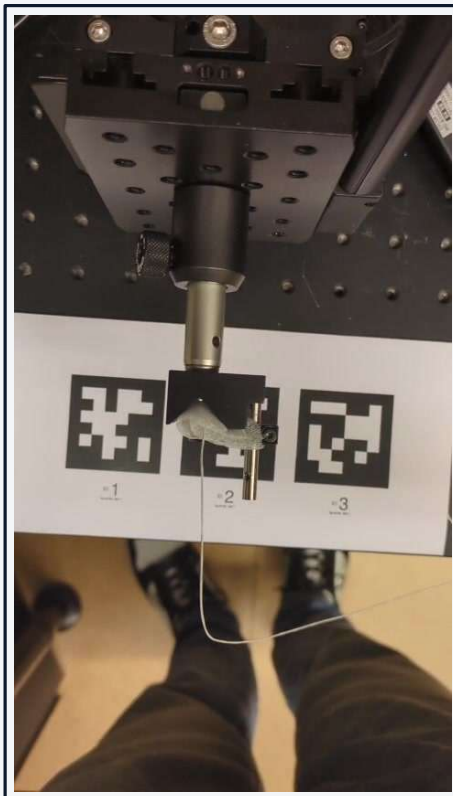
TOLERANCE

**5 pixels**

Minimal distance to target

IT WORKS · CLOSED LOOP

# The loop running end to end



► press play in PowerPoint to watch the full run

## WHAT YOU'RE SEEING

The camera starts off-centre; `center_tag.py --move` closes the loop on AprilTag ID 2 and the stages step in millimetres until the tag sits under the camera. This first end-to-end run uses the initial, untuned PI gains.

1

### Verify detection

`apriltag_tracker.py`

tag & angle look right

2

### Test motion

`kcube_raw_ethernet.py`

safe manual moves

3

### Dry-run

`center_tag.py`

corrections, no motion

4

### Go live

`center_tag.py --move`

closed-loop centering



BUILT TO RUN SAFELY

# Guardrails around the motion

## Software limits

Every axis is bounded 0–20 mm; positions are checked before any command is sent.

## All-or-nothing moves

For multi-axis moves, if any final position would be unsafe, none of the axes move.

## Step clamp

Each correction is capped at 5 mm, so a noisy reading can't cause a large lurch.

## Dry-run first

`center_tag.py` runs in preview mode by default — motion only with an explicit `--move` flag.

## Single owner of the link

Kinesis GUI must be closed; Python talks directly to the Ethernet TCP ports.

## Integral safety

The PI integral resets on tag loss, preventing wind-up while the target is missing.

## WHERE IT STANDS

# A working vision-to-motion loop

Detection, motion control, and the PI controller are integrated and demonstrated closing the loop on AprilTag 2.

## DELIVERED

- AprilTag detection with angle estimate
- Direct Ethernet control of 3 KDC101 stages
- PI centering loop, demonstrated live

## NEXT STEPS

- End effector creation
- Form labs printer testing and use
- Integration of new endoscopes